Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 32. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the test accuracy during training.

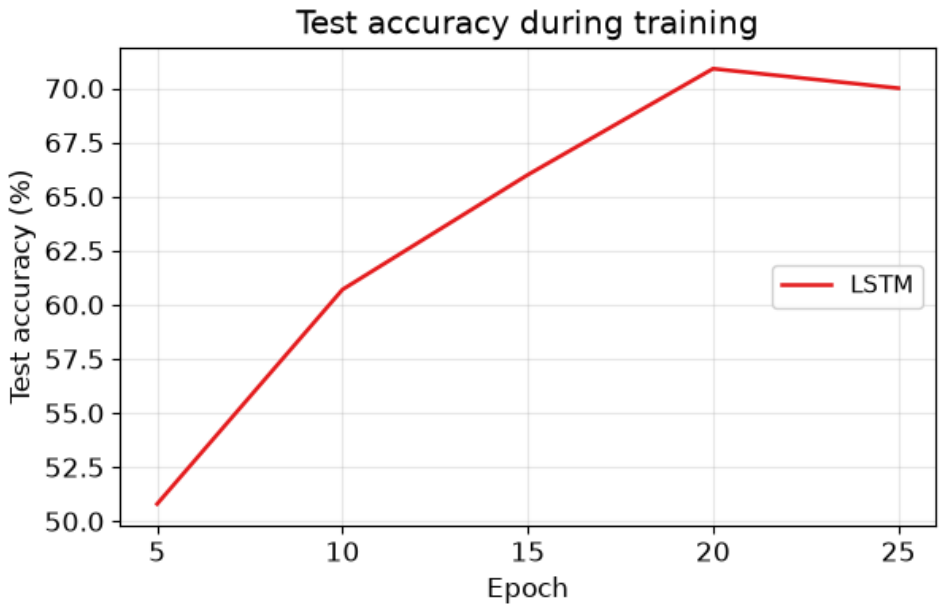


Figure 1: Test accuracy during training for the models in this experiment.

LSTM scores 20.1 points higher on 20 than on 5. LSTM scores 10.8 points higher on 15 than on 5.

Figure 2 shows the f1 score by dataset.

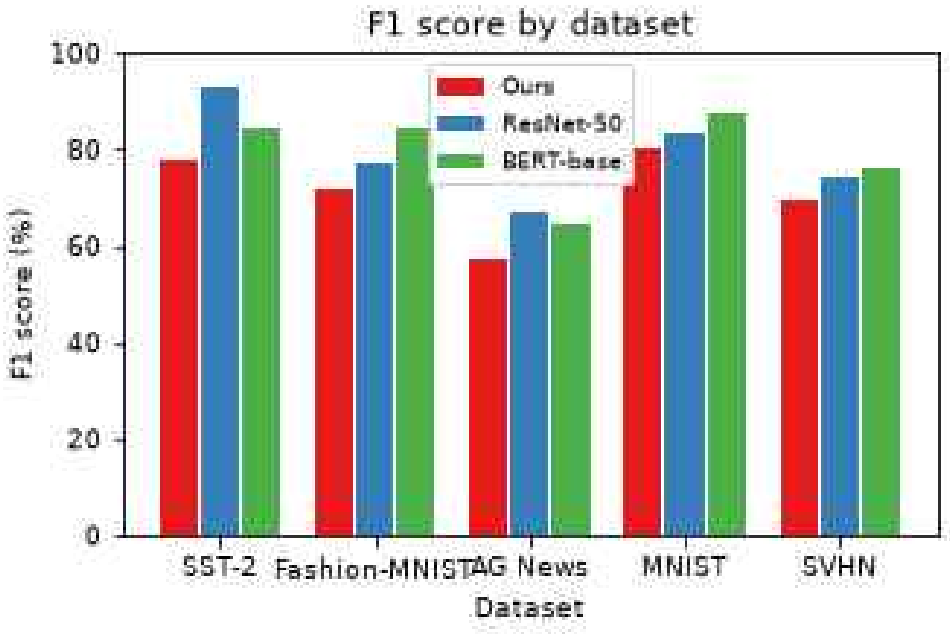


Figure 2: F1 score by dataset for the models in this experiment.

BERT-base achieves 26.1% F1 score on SVHN. BERT-base reaches 86.3% F1 score on Fashion-MNIST. ResNet-50 reaches 93.1% F1 score on SST-2. As shown in Figure 2, on SST-2, BERT-base is 7 points better than Ours.

3 Discussion

On STL-10, Naive Bayes obtains an accuracy of 83.3%. Overall, the results are in line with what we expected from the literature.